

From shop floors to showrooms: GenAI use cases unpacked

What makes GenAI a game-changer across the automotive value chain? Here is a snapshot of its real-world impact—beyond the buzzwords.

- **Dealerships get smarter:** A leading auto dealer leveraged GenAI directly on the showroom floor, not for design, but to automate service delivery, monitor employee performance, enhance customer experience, and improve security through video analytics. The result? Near-zero human supervision and fully automated operations.
- **Energy and utilities:** From predictive to prescriptive, GenAI in remote turbine ops doesn't just flag issues, it auto-generates work orders, schedules techs, checks parts, and suggests fixes from past data, transforming critical maintenance end-to-end.
- **Investment pulse:** When it comes to GenAI adoption, 14% of organisations are already investing, 19% are actively budgeting and testing, while 24% are in wait-and-watch mode. The rest remain undecided. However, one thing is clear: early adopters are already seeing returns, and others will soon follow suit.
- **GenAI-driven productivity gains:** GenAI is delivering significant productivity gains across key automotive functions: 37-39% in sales and customer service, 35-37% in maintenance operations, and 34-36% in production and assembly, with quality assurance also showing measurable improvements. This isn't just about efficiency, it is a clear sign of real, disruptive transformation.
- **Full-spectrum impact:** GenAI is reshaping the entire automotive value chain, from 3D modeling and EV optimisation in R&D to real-time quality checks in manufacturing, predictive diagnostics in services, and its cross-functional intelligence powering end-to-end innovation. With Industry 5.0 on the rise, we are enabling smarter connected production lines, predictive maintenance, energy monitoring, remote operations, and AR/VR-based worker training.

- Autonomous driving features, ranging from parking assist to fully self-driving modes

Here is where GenAI is beginning to shine, too. Think of systems that simulate thousands of traffic scenarios to train autonomous vehicles—all without a single real-world crash. Or in-car assistants that not only understand you but speak like you, adapting tone, language, and even emotion.

In the automobile sector, it is not just about speed or style anymore. It is about being smart. Vehicles that think, learn, predict, and even create. So the next time your car parks itself or your smart home anticipates your bedtime routine, take a second to appreciate what is happening under the hood. Because somewhere deep in the layers of data and code, your devices are learning to be more... you.

The global market shift is clear: Western Europe may have led the autonomous

vehicle charge, but Asia is firmly in the driver's seat now. Demand for autonomous and semi-autonomous vehicles across the region is rising. Projections show a compound annual growth rate (CAGR) of 12.1%. By 2027, the market could reach ₹1651.37 billion.

Connected cars are not just

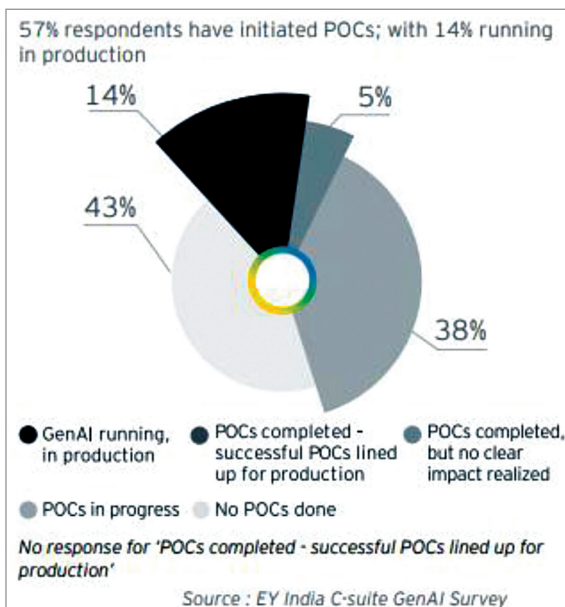


Fig. 3: Experience with GenAI POC engagements

about your vehicle talking to your phone. It is about cars communicating with other vehicles, devices, and infrastructure—essentially, becoming part of an intelligent network. With a projected CAGR of 17%, the connected car ecosystem is expected to reach 367 million units globally by 2027.

Automotive IoT has been building momentum since the early 2010s, but adoption has steepened dramatically in recent years. Driven by regulatory compliance, telematics, and usage-based insurance models, the market is expected to hit \$322 billion by 2028.

Why such growth? Because real-time data is king. Insurers are using IoT to assess claims based on actual driving behaviour. OEMs are embedding sensors everywhere—for predictive maintenance, performance tuning, and enhanced driver experience. Today's car is not just a vehicle—it is a digital companion. Auto manufacturers are racing to pack in rich in-vehicle experiences, infotainment systems, ADAS features, and more. It is no longer about horsepower or torque—it is about how connected, smart, and responsive the car feels.

Behind the scenes, smart factories are advancing, too. We have moved from Industry 4.0 to early signs of 5.0, where human-machine collaboration, sustainability, and personalised manufacturing are central.

The EV infrastructure challenge—and opportunity

The rise of EVs demands smarter infrastructure. IoT-enabled charging stations, predictive energy demand systems, real-time fleet charging optimisation, and battery analytics are becoming essential. Add to that vehicle-to-vehicle (V2V) communication and automated quality inspection—and we see a full ecosystem in motion.

Digitalisation is redefining